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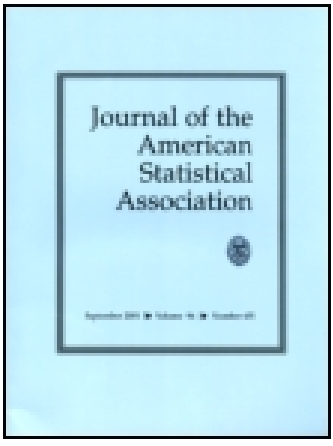

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Circular Error Probabilities

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CIRCULAR ERROR PROBABILITIES

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A problem which often arises in connection with the determination of probabilities of various miss distances of bombs and missiles is the following: Let x and y be two normally and independently distributed orthogonal components of the miss distance, each with mean zero and with standard deviations σ_x and σ_y, respectively, where for convenience one labels the components so that $\sigma_x \geq \sigma_y$. Now for various values of $c=\sigma_y/\sigma_x$, it is required to determine (1) the probability P that the point of impact lies inside a circle with center at the target and radius $K\sigma_x$, and (2) the value of K such that the probability is P that the point of impact lies inside such a circle. Solutions of (1), for $c=0.0(0.1)\ 1.0$ and $K=0.1\ (0.1)\ 5.8$, and (2), for the same values of c and $P=0.5$, 0.75, 0.9, 0.95, 0.975, 0.99, 0.995, 0.9975, and 0.999, are given along with some hypothetical examples of the application of the tables.

1. FORMULATION OF THE PROBLEM

This problem, which is essentially that of finding the probability that a quadratic form does not exceed a specified value, has been considered by a number of writers (see references). In particular, Grad and Solomon [6] have given formulas for $2k$ and $2k+1$ dimensions, together with brief four-place tables for two and three dimensions. For the two-dimensional case the following formulation, due to Fettis [3], is somewhat simpler.

The joint probability density function of two orthogonal variables x and y which are normally and independently distributed, each with mean zero and with standard deviations σ_x and σ_y, respectively, is given by

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} e^{-\frac{1}{2}[(x/\sigma_x)^2+(y/\sigma_y)^2]}. \tag{1}$$

The probability that a point (x, y), whose coordinates are chosen randomly and independently from this joint distribution, will lie within a circle with center at the origin and radius $K\sigma_x$ is

$$P(K, \sigma_x, \sigma_y) = \iint_{\sqrt{x^2+y^2}<K\sigma_x} f(x, y)dxdy. \tag{2}$$

If one introduces polar coordinates by letting $x/\sigma_x=\rho\cos\theta$ and $y/\sigma_x=\rho\sin\theta$, this probability takes the form

$$P(K, \sigma_x, \sigma_y) = \frac{\sigma_x}{2\pi\sigma_y}\int_0^{2\pi}\int_0^K e^{-\frac{1}{2}\rho^2[\cos^2\theta+(\sigma_x/\sigma_y)^2\sin^2\theta]}\rho d\rho d\theta. \tag{3}$$

If one lets $c=\sigma_y/\sigma_x$ (without loss of generality, one may ensure that $c\leq 1$ by calling the larger of the two standard deviations σ_x) and sets $2\theta=\phi$, one obtains

$$P(K, c) = \frac{1}{2\pi c}\int_0^{2\pi}\int_0^K e^{-(\rho^2/4c^2)[(1+c^2)-(1-c^2)\cos\phi]}\rho d\rho d\phi. \tag{4}$$

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If one then lets $z=\rho^2/4c^2$, (4) becomes

$$P(K, c) = \frac{2c}{\pi}\int_0^{\pi}\int_0^{K^2/4c^2} e^{-z[(1+c^2)-(1-c^2)\cos\phi]}dz d\phi. \tag{5}$$

Upon integration with respect to z, one obtains

$$P(K, c) = \frac{2c}{\pi}\int_0^{\pi} \frac{1 - e^{-(K^2/4c^2)[(1+c^2)-(1-c^2)\cos\phi]}}{(1+c^2)-(1-c^2)\cos\phi} d\phi. \tag{6}$$

It can easily be shown that equation (6) is equivalent to the result given by Grad and Solomon [6, equation (22)].

2. COMPUTATION OF THE TABLES

It was not found possible to perform in closed form the integration indicated in equation (6). Therefore it was necessary to resort to numerical integration, which was performed on the Burroughs E101-3 computer, employing the trapezoidal rule. Values of cos ϕ, accurate to eleven decimal places, were read into the machine by means of the tape input, while values of the exponential function, accurate to within a unit in the tenth decimal place, were computed as needed. The probability $P(K, c)$ was computed for $c=0.1(0.1)1.0$ and $K=0.1(0.1)5.8$. Eight decimal places were carried in the calculations, with error less than a unit in the seventh decimal place. The machine introduces a chopping error, since digits sufficiently far to the right of the decimal point are lost. If one or more digits are to be discarded, chopping may be replaced by rounding by the expedient of increasing by five the leftmost digit to be discarded and then chopping; this was done at one stage of the computation in order to avoid an accumulation of chopping errors when the integrands were summed as required by the trapezoidal rule. The use of the trapezoidal rule introduces a truncation error which depends on the interval $h(\phi)$. It was possible in most cases to obtain the required accuracy by using an interval $h_1(\phi)=\pi/20$ (radian) $=9°$. Smaller intervals were required, however, for high values of K combined with low values of c; specifically, the interval $h_2(\phi)=\pi/40$ (radian) $=4.5°$ was used for $c=0.2, K=4.0(0.1)5.8$ and for $c=0.1, K=1.2(0.1)3.5$, while the interval $h_3(\phi)=\pi/80$ (radian) $=2.25°$ was used for $c=0.1$, $K=3.6(0.1)5.8$. The results, rounded to seven decimal places, are shown in Table 1, which also includes values for $c=0$. The latter were obtained from a table of the normal probability function, since

$$P(K, 0) = 2\Phi(K) - 1 = (2\pi)^{-\frac{1}{2}}\int_{-K}^{K} e^{-x^2/2}dx. \tag{7}$$

It should be pointed out that if one sets $c=1$ in (6), the result is

$$P(K, 1) = 1 - e^{-K^2/2}. \tag{8}$$

Comparison of the eight-decimal-place results for $c=1$ with values found from (8) verifies that the error in the former is indeed less than a unit in the seventh decimal place.

In order to determine, for $c=0.1(0.1)0.9$, the value of K corresponding to

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TABLE 1. CIRCULAR ERROR PROBABILITIES $P(K, c)$

K \ c	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.1	.0796557	.0443987	.0242119	.0164176	.0123875	.0099377	.0082940	.0071157	.0062299	.0055400	.0049875
0.2	.1585194	.1339783	.0884533	.0628396	.0482413	.0390193	.0327123	.0281415	.0246824	.0219757	.0198013
0.3	.2358228	.2213804	.1739300	.1318281	.1039193	.0851535	.0719102	.0621386	.0546598	.0487639	.0440025
0.4	.3108435	.3010228	.2635181	.2139084	.1742045	.1451808	.1237982	.1076237	.0950495	.0850326	.0768837
0.5	.3829249	.3755884	.3481790	.3003001	.2532953	.2152886	.1857448	.1626829	.1443941	.1296286	.1175031
0.6	.4514938	.4457708	.4255605	.3846374	.3357384	.2914682	.2548177	.2251114	.2009797	.1811783	.1647298
0.7	.5160727	.5115048	.4960683	.4633258	.4170862	.3699305	.3280302	.2925654	.2629373	.2381583	.2172955
0.8	.5762892	.5725957	.5604457	.5349387	.4941882	.4474207	.4025628	.3627122	.3283453	.2989700	.2738510
0.9	.6318797	.6288721	.6191354	.5993140	.5651564	.5213998	.4759375	.4333628	.3953279	.3620135	.3330232
1.0	.6826895	.6802325	.6723586	.6568242	.6291249	.5900953	.5461319	.5025790	.4621421	.4257553	.3934693
1.1	.7286679	.7266597	.7202682	.7079681	.6859367	.6524489	.6116316	.5687467	.5272462	.4887873	.4539256
1.2	.7698607	.7682215	.7630305	.7532175	.7359558	.7079973	.6714269	.6306168	.5893494	.5498736	.5132477
1.3	.8063990	.8050648	.8008554	.7929968	.7793550	.7567265	.7249673	.6873122	.6474394	.6079822	.5704426
1.4	.8384867	.8374049	.8340018	.8277048	.8169851	.7989288	.7720889	.7383089	.7007900	.6623035	.6246889
1.5	.8663856	.8655127	.8627728	.8577362	.8493071	.8350816	.8129287	.7833962	.7489500	.7122546	.6753475
1.6	.8904014	.8897008	.8875060	.8834914	.8768644	.8657559	.8478393	.8226246	.7917194	.7574708	.7219627
1.7	.9108691	.9103102	.9085619	.9053766	.9001746	.8915536	.8773116	.8562471	.8291137	.7977882	.7462539
1.8	.9281394	.9276964	.9263125	.9237989	.9197275	.9130680	.9019110	.8846624	.8613238	.8332175	.8021013
1.9	.9425669	.9422182	.9411299	.9391586	.9359855	.9308615	.9222277	.9083609	.8886731	.8639149	.8355255
2.0	.9544997	.9542272	.9533775	.9518415	.9493815	.9454546	.9388418	.9278799	.9115762	.8901495	.8646647
2.1	.9642712	.9640598	.9634011	.9622127	.9603170	.9573205	.9522999	.9437668	.9305013	.9122714	.8897495
2.2	.9721931	.9720304	.9715237	.9706109	.9691597	.9668845	.9631017	.9565522	.9459386	.9306821	.9110784
2.3	.9785518	.9784275	.9780408	.9773450	.9762419	.9745239	.9716934	.9667306	.9583739	.9458085	.9289946
2.4	.9836049	.9835108	.9832180	.9826918	.9818594	.9805703	.9784661	.9747495	.9682698	.9580804	.9438652
2.5	.9875807	.9875100	.9872900	.9868953	.9862720	.9853112	.9837569	.9810035	.9760522	.9679136	.9560631
2.6	.9906776	.9906249	.9904612	.9901674	.9897045	.9889934	.9878527	.9858331	.9821023	.9756969	.9659525
2.7	.9930661	.9930271	.9929062	.9926894	.9923483	.9918260	.9909944	.9895268	.9867530	.9817837	.9738786
2.8	.9948897	.9948612	.9947727	.9946141	.9943649	.9939842	.9933821	.9923249	.9902888	.9864876	.9801589
2.9	.9962684	.9962477	.9961834	.9960684	.9958878	.9956126	.9951798	.9944246	.9929482	.9900803	.9850792
3.0	.9973002	.9972853	.9972391	.9971564	.9970266	.9968294	.9965205	.9959854	.9949274	.9927925	.9888910

$P(K, c)$ = the probability that a point falls inside a circle whose center is at the origin and whose radius is K times the larger standard deviation, c being the ratio of the smaller standard deviation to the larger standard deviation.

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TABLE 1— (*cont.*)

K \ c	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
3.1	.9980648	.9980542	.9980212	.9979622	.9978699	.9977296	.9975109	.9971348	.9963851	.9948168	.9918113
3.2	.9986257	.9986182	.9985949	.9985533	.9984880	.9983892	.9982356	.9979733	.9974478	.9963105	.9940240
3.3	.9990332	.9990279	.9990116	.9989824	.9989368	.9988677	.9987607	.9985792	.9982147	.9974004	.9956822
3.4	.9993261	.9993225	.9993112	.9992909	.9992593	.9992115	.9991376	.9990129	.9987626	.9981868	.9969113
3.5	.9995347	.9995323	.9995245	.9995105	.9994888	.9994559	.9994053	.9993204	.9991502	.9987480	.9978125
3.6	.9996818	.9996801	.9996748	.9996653	.9996505	.9996281	.9995938	.9995364	.9994218	.9991442	.9984662
3.7	.9997844	.9997832	.9997797	.9997733	.9997633	.9997482	.9997251	.9996867	.9996102	.9994208	.9989352
3.8	.9998553	.9998545	.9998522	.9998478	.9998412	.9998311	.9998157	.9997902	.9997396	.9996119	.9992682
3.9	.9999038	.9999033	.9999018	.9998989	.9998945	.9998878	.9998776	.9998608	.9998276	.9997426	.9995020
4.0	.9999367	.9999363	.9999353	.9999334	.9999305	.9999261	.9999195	.9999085	.9998870	.9998309	.9996645
4.1	.9999587	.9999585	.9999578	.9999566	.9999547	.9999519	.9999475	.9999404	.9999266	.9998900	.9997763
4.2	.9999733	.9999732	.9999727	.9999720	.9999707	.9999689	.9999661	.9999616	.9999527	.9999292	.9998523
4.3	.9999829	.9999828	.9999826	.9999821	.9999813	.9999801	.9999783	.9999754	.9999698	.9999548	.9999034
4.4	.9999892	.9999891	.9999889	.9999886	.9999881	.9999874	.9999863	.9999845	.9999809	.9999715	.9999375
4.5	.9999932	.9999932	.9999931	.9999929	.9999925	.9999921	.9999914	.9999902	.9999881	.9999822	.9999599
4.6	.9999958	.9999957	.9999957	.9999955	.9999954	.9999951	.9999947	.9999939	.9999926	.9999889	.9999746
4.7	.9999974	.9999974	.9999973	.9999973	.9999971	.9999970	.9999967	.9999963	.9999955	.9999932	.9999840
4.8	.9999984	.9999984	.9999984	.9999983	.9999983	.9999982	.9999980	.9999977	.9999972	.9999959	.9999901
4.9	.9999990	.9999990	.9999990	.9999990	.9999990	.9999989	.9999988	.9999986	.9999983	.9999975	.9999939
5.0	.9999994	.9999994	.9999994	.9999994	.9999994	.9999993	.9999993	.9999992	.9999990	.9999985	.9999963
5.1	.9999997	.9999997	.9999997	.9999996	.9999996	.9999996	.9999996	.9999995	.9999994	.9999991	.9999978
5.2	.9999998	.9999998	.9999998	.9999998	.9999998	.9999998	.9999998	.9999997	.9999997	.9999995	.9999987
5.3	.9999999	.9999999	.9999999	.9999999	.9999999	.9999999	.9999999	.9999998	.9999998	.9999997	.9999992
5.4	.9999999	.9999999	.9999999	.9999999	.9999999	.9999999	.9999999	.9999999	.9999999	.9999998	.9999995
5.5	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	.9999999	.9999999	.9999999	.9999997
5.6								1.0000000	1.0000000	.9999999	.9999998
5.7										1.0000000	.9999999
5.8											1.0000000
5.9											
6.0											

$P(K, c)$ = the probability that a point falls inside a circle whose center is at the origin and whose radius is K times the larger standard deviation, c being the ratio of the smaller standard deviation to the larger standard deviation.

$P=0.5$, 0.75, 0.9, 0.95, 0.975, 0.99, 0.995, 0.9975, and 0.999, it was necessary to interpolate inversely in the unrounded table of $P(K, c)$. These computations were also performed on the E101-3, using Aitken's method of interpolation with a tolerance of 5×10^{-8} and with provision for up to nine-point interpolation if the tolerance is not met for fewer points. In practice, the tolerance was never met, so that nine-point interpolation was always used; even so, from these results it was possible to determine five-decimal-place values of K only for small values of P (0.5 always, 0.75 usually, and 0.9 occasionally). Where this was not possible, direct interpolation was performed to determine P for the best estimate of K that could be obtained from the initial inverse interpolation; then the inverse interpolation was repeated, using the estimated K and the corresponding P as coordinates of the first point and discarding the ninth point of the initial inverse interpolation. In this manner, it was possible to determine the required values of K, accurate to within a unit in the fifth decimal place. The results, rounded to five decimal places, are shown in Table 2, along with results for $c=0$, obtained by inverse interpolation in a table of the normal probability function, and results for $c=1$, obtained by substituting numerical values of P in the equation

$$K = \sqrt{-2\ln(1-P)}, \tag{9}$$

which was found by solving (8) for K.

3. COMPARISON WITH OTHER TABLES

Grad and Solomon [6, Table I] have tabulated four-decimal-place values of the probability $P(a_1x_1^2+a_2x_2^2<t)$, where x_1 and x_2 are normally and independently distributed with zero means and unit variances, for eight selected pairs of values of a_1 and a_2 such that $a_2\geq a_1$, $a_1+a_2=1$ and for $t=0.1(0.1)1.0(0.5)2.0(1.0)5.0$. Table 1 of the present paper gives seven-decimal-place values of the probability $P(cx_1^2+x_2^2<K^2)$, where again x_1 and x_2 are normally and independently distributed with zero means and unit variances, for $c=0.0(0.1)1.0$ and $K=0.0(0.1)5.8$. The two probabilities are equivalent if one sets

$$c = \sqrt{\frac{a_1}{a_2}}, \qquad K = \sqrt{\frac{t}{a_2}}.$$

The pairs of values a_2, a_1 and the corresponding values of c are as follows:

a_2, a_1	.5, .5	.6, .4	.7, .3	.8, .2	.9, .1	.95, .05	.99, .01	1, 0
c	1.0000	0.8165	0.6547	0.5000	0.3333	0.2294	0.1005	0.0000

Table 1 of the present paper has eleven values of c spanning (0, 1) with a uniform interval of 0.1. The maximum value of t for the Grad-Solomon tables is 5.0, corresponding to a maximum value of K ranging from $\sqrt{5}\doteq2.236$ for $a_2=1$ to $\sqrt{10}\doteq3.162$ for $a_2=.5$; in Table 1 of the present paper, values of K go up to 5.8 unless, for some lower value of K, the probability, when rounded to seven decimal places, becomes 1.0000000. Thus Table 1 of the present paper is considerably more extensive (621 entries as compared with 120) than the Grad-

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TABLE 2. VALUES OF K CORRESPONDING TO CUMULATIVE PROBABILITY P

P \ c	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
.5000	0.67449	0.68199	0.70585	0.74993	0.80785	0.87042	0.93365	0.99621	1.05769	1.11807	1.17741
.7500	1.15035	1.15473	1.16825	1.19246	1.23100	1.28534	1.35143	1.42471	1.50231	1.58271	1.66511
.9000	1.64485	1.64791	1.65731	1.67383	1.69918	1.73708	1.79152	1.86253	1.94761	2.04236	2.14597
.9500	1.95996	1.96253	1.97041	1.98420	2.00514	2.03586	2.08130	2.14598	2.23029	2.33180	2.44775
.9750	2.24140	2.24365	2.25053	2.26255	2.28073	2.30707	2.34581	2.40356	2.48494	2.58999	2.71620
.9900	2.57583	2.57778	2.58377	2.59421	2.60995	2.63257	2.66533	2.71515	2.79069	2.89743	3.03485
.9950	2.80703	2.80883	2.81432	2.83289	2.83830	2.85894	2.88859	2.93347	3.00431	3.11073	3.25525
.9975	3.02334	2.02500	3.03010	3.03898	3.05234	3.07144	3.09871	3.13969	3.20586	3.31099	3.46164
.9990	3.29053	3.29206	3.29673	3.30489	3.31715	3.33464	3.35949	3.39647	3.45698	3.55939	3.71692

Solomon table, as well as much more accurate (seven decimal places as compared with four).

Grad and Solomon have no table analogous to Table 2 of the present paper, which gives the value of K corresponding to cumulative probability P for $c=0.0(0.1)1.0$ and $P=0.5$, 0.75, 0.9, 0.95. 0.975. 0.99, 0.995, 0.9975, and 0.999.

Grad and Solomon [6, Table II] have also tabulated four-decimal-place values of the spherical error probability $P(a_1x_1^2+a_2x_2^2+a_3x_3^2<t)$ for nine selected combinations of a_3, a_2, and a_1 and for the same values of t as in their Table I.

Solomon [14] recently prepared more extensive and more accurate tables for both the two and three-dimensional cases, including tables analogous to Table 2 of the present paper.

The Numerical Analysis Department of the RAND Corporation [9] and Burington and May [1, pp. 102–5] have published tables of offset circle probabilities for the case $\sigma_x=\sigma_y=\sigma(c=1$ or $a_2=a_1=0.5)$, which give the probability of missing (or hitting) a circle of radius r_d/σ whose center is at a distance D/σ from the origin. DiDonato and Jarnagin[1] recently published extensive tables of offset ellipse probabilities.

4. APPLICATIONS

The applications of the theory to the determination of hit probabilities for circular targets is well known (see, for example, papers by Scheffé [13], Fraser [4, 5], and Grad and Solomon [6]). A few somewhat oversimplified examples will be given, however, for the purpose of illustrating the use of the Tables. The values assigned to the parameters in these examples do not represent the capabilities of any actual weapon.

Example 1. What is the probability that the miss distance of a bomb is less than 180 feet if there are no systematic errors and if the random errors in two orthogonal directions are normally and independently distributed with standard deviations $\sigma_x=120$ feet and $\sigma_y=72$ feet? Solution: One has $c=\sigma_y/\sigma_x=72/120=0.6$ and $K\sigma_x=180$, so that $K=180/120=1.5$. By reference to Table 1, one finds the required probability $P(1.5, 0.6)=0.8129287$.

Example 2. An ICBM has range errors and lateral errors which are normally and independently distributed with standard deviations $\sigma_x=2$ miles and $\sigma_y=1$ mile, respectively. What is the radius of a circle which may be expected to include (a) 50% (b) 95% of the points of impact? Solution: One has $c=\sigma_y=\sigma_x=1/2=0.5$. (a) By reference to Table 2, one finds that when $c=0.5$ and $P=0.5$, $K=0.87042$. Hence the radius of a circle containing 50% of the points of impact (called the circular probable error) is $K\sigma_x=0.87042(2)=1.74084$ miles. (b) For $c=0.5$ and $P=0.95$, $K=2.03586$. Hence the radius of a circle containing 95% of the points of impact is $2.03586(2)=4.07172$ miles.

Example 3. In high altitude bombing, the range error and the lateral error are normally and independently distributed with standard deviations $\sigma_x=1000$

[1] DiDonato, A. R. and Jarnagin, M. P., "Integration of the General Bivariate Gaussian Distribution over an Offset Ellipse," Report No. 1710, U. S. Naval Weapons Laboratory, Dahlgren, Virginia, 1960.

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feet and $\sigma_y = 450$ feet, respectively. Assuming no systematic errors, what is the probability that the miss distance is less than one-quarter mile (1320 feet)? Solution: One has $c = \sigma_y/\sigma_x = 450/1000 = 0.45$ and $K = 1320/1000 = 1.32$. Double interpolation gives the following values for $P(1.32, 0.45)$: 0.7760240 (linear); 0.7776033 (three-point Lagrangian); 0.7776141 (four-point Lagrangian). Thus, in this example, the results for three- and four-point Lagrangian interpolation agree to four decimal places, with a discrepancy of just over a unit in the fifth decimal place, while the result of linear interpolation disagrees with these by almost two units in the third decimal place. For practical purposes, three-point Lagrangian interpolation will almost always give satisfactory results, though linear interpolation often will not.

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